

UTXO Ownership and Fee Settlement in the Lux 9-Chain Topology

The X-Native Fee Rail: P/X as Money and Stake,
the Rest as Metered Execution Lanes

Zach Kelling

Lux Industries / Lux Network
research@lux.network @zeekay

LP-companion spec • Version 0.1 (Draft) • Aligns to LP-134 / LP-7050

Abstract

The canonical Lux topology is nine orthogonal chains — P C X Q Z A B M F (LP-134). This note fixes the *economic* invariants that keep that topology clean: which chains own canonical asset state, and how fees flow. The rule is deliberately narrow. Only P and X are canonical UTXO state machines: P is the validator/stake/control-plane ledger and X is the asset/settlement/payment ledger. X is additionally the canonical fee asset. Every other chain (C D Q Z A B M F) is a domain-specific execution lane that holds VM-local state plus an X-funded fee balance, meters its own work, and settles rewards and remainders back to X (with validator economics accounted on P). We give the fee-funding model (per-user X import/export plus per-chain fee-root settlement, finalized on Q), a per-chain fee policy, and seven conservation invariants that prevent any lane from silently becoming a second asset ledger. Shielded state on Z and encrypted balances on F are UTXO-like in form but are explicitly *not* the canonical transparent X set — they are commitments over X-backed assets and must prove conservation against it.

Contents

1	The invariant: only P and X own UTXOs	2
2	How the other chains relate to P/X	2
3	The fee model	3
3.1	Option 1 — X-native fee import (sovereign clarity)	3
3.2	Option 2 — fee-root settlement (throughput)	3
3.3	Recommended hybrid	3
4	Per-chain fee policy	3
5	UTXO-like $\neq$ canonical UTXO: Z and F	4
6	Fee & conservation invariants	4
7	Summary	5

1 The invariant: only P and X own UTXOs

A recurring failure mode in multi-chain systems is that specialized chains quietly accrete the powers of a money ledger — they mint balances, hold custody of value, and become a second, unreconciled source of asset truth. Lux forecloses this by convention:

Only P and X are canonical UTXO state machines. No other chain becomes an asset ledger unless it imports/exports value through X.

P-Chain — stake / control plane. P holds UTXO-style state for validator stake, delegation stake, validator-set transitions, staking rewards, sovereign-network / L1 registration, and chain-creation / network-governance control transactions. P is the *stake and reward* rail.

X-Chain — assets / settlement. X holds UTXO-style state for native assets, transfers, settlement, bridge lock/release settlement, DEX deposit/withdraw settlement, and cross-chain import/export accounting. X is the *money* rail and the canonical fee asset.

Everything else consumes P/X settlement or maintains its own VM-local domain state — never a canonical asset UTXO set of its own.

2 How the other chains relate to P/X

Chain	State kind	Relation to P/X
C	Account / EVM	Fees paid in EVM gas inside C; canonical asset movement imports/exports through X.
D	DEX / matching	Funds enter from X via ImportTx/deposit, trade inside D, withdraw/export back to X.
Q	Finality / checkpoint	Not a user-asset UTXO chain. Fees systemic/metered; fee accounting anchors to X/P. Finality must never be hostage to app fee markets.
Z	ZK / proof / rollup	Commitments/nullifiers are UTXO- <i>like</i> but <i>not</i> the canonical transparent X set. Public settlement roots back to X.
A	AI / attestation / receipts	Not UTXO. Pays via X or an imported fee balance; workers settle to X.
B	Bridge / teleport state machine	Not canonical UTXO. Owns bridge message lifecycle; locked/released assets settle on X. Consumes M attestations.
M	MPC / custody / signing / attestation	Not UTXO. Signs and attests; holds no canonical asset UTXOs except X-backed references.
F	FHE / confidential compute	Encrypted state / ciphertext commitments. Not canonical UTXO. Settles to X.

Simple rule. P/X have UTXOs. Every other chain has domain state plus X-backed fee balances.

3 The fee model

X is the canonical fee asset and payment rail. Each chain meters its own execution. Fees are paid from local imported balances or settled back to X. P handles validator economics and reward distribution.

So X = money rail, P = staking/reward rail, and C D A B M F Q Z are execution lanes that consume fees.

3.1 Option 1 — X-native fee import (sovereign clarity)

A user imports a fee budget from X into the target chain; execution charges the local balance; any remainder can be exported back:

X ExportTx → target ImportTx → local fee balance → execution charge → (optional) export remainder to X.

Clean for D F A B M: they meter domain-specific work without mutating X on every operation.

Example (FHE job on F). (1) user holds LUX on X; (2) exports a 10-LUX budget X→F; (3) F imports it as local fee credit; (4) GPU workers execute the encrypted job; (5) F pays workers/validators from the local pool; (6) remaining credit exports back to X.

3.2 Option 2 — fee-root settlement (throughput)

Each chain accumulates fees internally and periodically commits a fee settlement root to X:

local fee accounting → epoch fee root → X settlement / distribution.

Better for high-throughput D/F/A where per-tx X imports would be too heavy.

3.3 Recommended hybrid

Use both. User-facing funding via X import/export into local balances; high-throughput settlement via per-chain fee roots periodically settled to X/P; validator/operator rewards distributed by P using *finalized* fee roots. This yields: X = source of fee currency; local chain = execution metering; Q = finality of fee roots; P = validator reward / stake accounting.

4 Per-chain fee policy

Chain	Fees	Paid from / settlement
P	staking tx, validator registration, chain/subnet creation	P UTXO; rewards to validators/delegators on P.
X	transfer, asset creation, import/export	X UTXO; the canonical settlement + fee asset.
C	EVM gas	C balance imported/bridged from X; optional X-backed native accounting.
D	order place/cancel, taker/maker, settlement	D local balance funded from X; assets enter/exit via X Import/Export.

Chain	Fees	Paid from / settlement
Q	cert/checkpoint/finality inclusion	system budget or local Q balance funded from X. Keep cheap/systemic; do not make finality hostage to app fee markets.
Z	proof verification, rollup batch inclusion, private settlement/nullifier processing	Z balance funded from X; roots/public outputs settle to X/Q.
A	inference receipt, model registry update, attestation verify, PoAI job	A balance funded from X; AI workers paid from A pool, settled to X.
B	bridge message, route, attestation inclusion, relayer	B balance funded from X or deducted from bridged amount; lock/release/mint/burn reference X assets.
M	MPC signing session, custody approval, bridge attestation	M balance funded from X; MPC signers paid from M pool, settled to X.
F	FHE job, encrypted-state update, GPU compute, threshold decrypt	F balance funded from X; FHE/GPU workers + decrypt committee paid from F pool, settled to X.

5 UTXO-like $\neq$ canonical UTXO: Z and F

Z may carry commitments, nullifiers, shielded notes, and rollup exits; F may carry encrypted balances and ciphertext state. In *form* these resemble a UTXO set, but architecturally they are **not** the canonical transparent X asset ledger:

X = canonical transparent UTXO assets, Z/F = shielded / encrypted commitments over X-backed assets.

Consequently Z and F private state must *prove conservation* against X-backed commitments (a shielded or encrypted balance may never exceed, or mint outside, the X value it represents).

6 Fee & conservation invariants

- I1. No chain may mint canonical asset supply except through X/P defined rules.
- I2. Every non-P/X fee balance must be funded by an X/P import, a genesis allocation, or a finalized fee settlement.
- I3. D/F/A/M/B worker rewards must be claimable / settled back to X.
- I4. Q finality fees cannot block safety or finality.
- I5. Z/F private state must prove conservation against X-backed commitments.
- I6. Bridge fees on B are message/relayer fees — not arbitrary custody of canonical assets.
- I7. M signs attestations but never becomes the asset ledger.

7 Summary

UTXO chains: P = staking/validator UTXO; X = asset/settlement UTXO. **Fees:** X is the canonical fee asset rail; P handles staking/reward economics; every other chain meters locally with fee balances funded from X, then settles/rewards back to X/P through finalized roots (finalized on Q). That keeps the nine-chain architecture clean: P/X are the money/stake ledgers, and C D Q Z A B M F are specialized execution lanes. It is the same discipline as the T-braid removal: separate what a primitive *is* (asset value, threshold crypto) from *where* it is applied (an execution lane) — values, not places.